

Q1. Equal and Opposite Forces

By Newton's Third Law the two pulls are equal and opposite: $|\vec{\mathbf{F}}_{A \rightarrow B}| = |\vec{\mathbf{F}}_{B \rightarrow A}|$.

Q2. The Zero-Force Point

(1) On the line AB the forces on m are collinear and opposite, so they cancel when

$$\frac{GM_A m}{x^2} = \frac{GM_B m}{(D-x)^2} \implies \frac{M_A}{x^2} = \frac{M_B}{(D-x)^2}.$$

(2) Taking square roots and solving:

$$x = D \frac{\sqrt{M_A}}{\sqrt{M_A} + \sqrt{M_B}} = (5.0 \times 10^{12}) \frac{2 \times 10^{15}}{3 \times 10^{15}} = \boxed{3.3 \times 10^{12} \text{ m}}.$$

Q3. Orbital Period

Let $r_A + r_B = D$ be the orbital radii about the common centre of mass, with $\omega = 2\pi/T$. Equating the mutual force to the centripetal force for each star,

$$\frac{GM_A M_B}{D^2} = M_A \omega^2 r_A = M_B \omega^2 r_B.$$

With $M_A r_A = M_B r_B$ and $r_A + r_B = D$, we get $r_A = \frac{M_B D}{M_A + M_B}$; substituting and using $\omega = 2\pi/T$ gives the modified Kepler's Third Law:

$$\boxed{T^2 = \frac{4\pi^2 D^3}{G(M_A + M_B)}}.$$

Numerically, with $M_A + M_B = 5.0 \times 10^{30} \text{ kg}$:

$$T = 2\pi \sqrt{\frac{(5.0 \times 10^{12})^3}{(6.67 \times 10^{-11})(5.0 \times 10^{30})}} = \boxed{3.8 \times 10^9 \text{ s}} \approx 122 \text{ yr}.$$

Q4. Orbital Speeds

Same angular speed means $v_A/v_B = r_A/r_B = M_B/M_A$, hence

$$\boxed{M_A v_A = M_B v_B}.$$